

CYBR 2300 – Week 4 Homework: IP Addressing, Subnetting, and CIDR (50 points)

Name: __Connor Okey__ Date: __9/15/2026__

IP Address Classes (Quick Reference)

Class Ranges

Class	First Octet Range	Description
Class A	1 – 127	Large networks
Class B	128 – 191	Medium networks
Class C	192 – 223	Small networks
Class D	224 – 239	Multicast (not used for hosts)
Class E	240 – 255	Experimental

Note:

- 127.x.x.x is reserved for **loopback (testing)**

Default Subnet Masks

Class Default Subnet Mask

Class A 255.0.0.0

Class B 255.255.0.0

Class C 255.255.255.0

Network vs Host Structure

Class	Network Portion	Host Portion
Class A	1st octet	Last 3 octets
Class B	First 2 octets	Last 2 octets
Class C	First 3 octets	Last octet

Binary Leading Bits (Advanced)

Class	Leading Bits
Class A	0
Class B	10
Class C	110

Private IP Address Ranges

Range	Use
10.0.0.0 – 10.255.255.255	Private (large networks)
172.16.0.0 – 172.31.255.255	Private (medium networks)
192.168.0.0 – 192.168.255.255	Private (home networks)

Simple Way to Remember

- 1-127 → Class A → Big
- 128-191 → Class B → Medium
- 192-223 → Class C → Small

1. IP Address Classes (FAST WAY)

Look at the **first number (first octet)**:

- 1-127 → Class A
 - 128-191 → Class B
 - 192-223 → Class C
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2. Network vs Host (FAST WAY)

Use the class:

- Class A → N.H.H.H
- Class B → N.N.H.H
- Class C → N.N.N.H

👉 Example:

192.168.1.10 → Class C

→ Network = 192.168.1

→ Host = 10

3. Network Address (FAST WAY)

👉 Rule: **Turn all host parts to 0**

- 192.168.1.25 /24 → **192.168.1.0**
- 10.10.48.80 /16 → **10.10.0.0**

👉 If you get stuck:

“Network = keep network, zero out host”

4. Host Address (FAST WAY)

👉 Rule: **Remove the network part**

- 192.168.1.25 /24 → Host = **0.0.0.25**

- 10.10.48.80 /16 → Host = **0.0.48.80**

5. Default Subnet Masks (MEMORIZE THIS)

- Class A → **255.0.0.0**
- Class B → **255.255.0.0**
- Class C → **255.255.255.0**

👉 Shortcut:

- A = 1 block
- B = 2 blocks
- C = 3 blocks

6. CIDR & Subnetting (SUPER FAST METHOD)

Common Ones to Memorize

CIDR	Subnet Mask	Hosts
/24	255.255.255.0	254
/25	255.255.255.128	126
/26	255.255.255.192	62
/27	255.255.255.224	30
/28	255.255.255.240	14

Host Formula (EXAM RULE)

👉 Hosts = $2^n - 2$

- /24 → 8 bits → $2^8 - 2 = 254$
 - /26 → 6 bits → $2^6 - 2 = 62$
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Block Size Trick (VERY IMPORTANT)

👉 Block size = **256 - subnet number**

- $/26 \rightarrow 192 \rightarrow 256 - 192 = \mathbf{64}$
- $/27 \rightarrow 224 \rightarrow 256 - 224 = \mathbf{32}$

👉 That tells you subnet jumps:

- $/26 \rightarrow 0, 64, 128, 192$
 - $/27 \rightarrow 0, 32, 64, 96...$
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Part 1: IP Address Classes (10 points)

Identify the class (A, B, C, D, E).

10.250.1.1 → Class: _A_____

150.10.15.0 → Class: _B_____

192.14.2.0 → Class: _C_____

148.17.9.1 → Class: _B_____

193.42.1.1 → Class: _C_____

126.8.156.0 → Class: _A_____

220.200.23.1 → Class: _C_____

230.230.45.58 → Class: _D_____

177.100.18.4 → Class: _B_____

119.18.45.0 → Class: _A_____

249.240.80.78 → Class: _E_____

199.155.77.56 → Class: _C_____

117.89.56.45 → Class: _A_____

215.45.45.0 → Class: _C_____

199.200.15.0 → Class: _C_____

95.0.21.90 → Class: _A_____

33.0.0.0 → Class: _A_____

158.98.80.0 → Class: _B_____

219.21.56.0 → Class: _C_____

Part 2: Network vs Host (10 points)

Identify the NETWORK and HOST portion.

177.100.18.4

Network: _177.100_____

Host: ____18.4_____

119.18.45.0

Network: _119_____

Host: _18.45.0_____

209.240.80.78

Network: _209.240.80_____

Host: _78_____

199.155.77.56

Network: _199.155.77____

Host: _56_____

117.89.56.45

Network: _117_____

Host: _89.56.45__

215.45.45.0

Network: _215.45.45_____

Host: _0_____

192.200.15.0

Network: _192.200.15_____

Host: _0_____

95.0.21.90

Network: _95__

Host: _0.21.90_____

33.0.0.0

Network: _33_____

Host: _0.0.0_____

158.98.80.0

Network: _158.98_____

Host: __80.0_____

217.21.56.0

Network: _217.21.56_____

Host: _0_____

10.250.1.1

Network: _10_____

Host: __250.1.1_____

150.10.15.0

Network: __150.10_____

Host: ____15.0__

192.14.2.0

Network: _192.14.2_____

Host: ____0_____

148.17.9.1

Network: _148.17_____

Host: ____9.1_____

193.42.1.1

Network: _193.42.1_____

Host: __1__

126.8.156.0

Network: __126__

Host: __8.156.0__

220.200.23.1

Network: __220.200.23__

Host: __1__

Part 3: Network Address (10 points)

Find the network address.

IP: 188.10.18.2 Mask: 255.255.0.0

Network Address: __: 188.10.0.0__

IP: 10.10.48.80 Mask: 255.255.255.0

Network Address: ____10.10.48.0____

IP: 192.149.24.191 Mask: 255.255.255.0

Network Address: __192.149.24.0____

IP: 150.203.23.19 Mask: 255.255.0.0

Network Address: __150.203.0.0____

IP: 10.10.10.10 Mask: 255.0.0.0

Network Address: ____10.0.0.0____

IP: 186.13.23.110 Mask: 255.255.255.0

Network Address: __186.13.23.0____

IP: 223.69.230.250 Mask: 255.255.0.0

Network Address: __223.69.0.0____

IP: 200.120.135.15 Mask: 255.255.255.0
Network Address: _200.120.135.0_____

IP: 27.125.200.151 Mask: 255.0.0.0
Network Address: _27.0.0.0_____

IP: 199.20.150.35 Mask: 255.255.255.0
Network Address: ___199.20.150.0_____

IP: 191.55.165.135 Mask: 255.255.255.0
Network Address: ___191.55.165.0_____

IP: 28.212.250.254 Mask: 255.255.0.0
Network Address: ___28.212.0.0_____

Part 4: Host Address (5 points)

Find the host portion.

IP: 188.10.18.2 Mask: 255.255.0.0
Host Address: ___0.0.18.2_____

IP: 10.10.48.80 Mask: 255.255.255.0
Host Address: ___0.0.0.80_____

IP: 222.49.49.11 Mask: 255.255.255.0
Host Address: _____0.0.0.11____

IP: 128.23.230.19 Mask: 255.255.0.0
Host Address: ___0.0.230.19_____

IP: 10.10.10.10 Mask: 255.0.0.0
Host Address: ___0.10.10.10_____

IP: 200.113.123.11 Mask: 255.255.255.0

Host Address: ___0.0.0.11_____

Part 5: Default Subnet Masks (5 points)

Write the default subnet mask.

177.100.18.4 → Subnet Mask: ___255.255.0.0_____

119.18.45.0 → Subnet Mask: ___255.0.0.0_____

191.249.234.191 → Subnet Mask: ___255.255.0.0___

223.23.223.109 → Subnet Mask: ___255.255.255.0_____

10.10.250.1 → Subnet Mask: ___255.0.0.0__

126.123.23.1 → Subnet Mask: ___255.255.0.0_____

223.69.230.250 → Subnet Mask: ___255.255.255.0___

192.12.35.105 → Subnet Mask: ___255.255.255.0___

77.251.200.51 → Subnet Mask: ___255.0.0.0___

189.210.50.1 → Subnet Mask: ___255.255.0.0__

88.45.65.35 → Subnet Mask: ___255.0.0.0___

128.212.250.254 → Subnet Mask: ___255.255.0.0_____

193.100.77.83 → Subnet Mask: ___255.255.255.0__

125.125.250.1 → Subnet Mask: ___255.0.0.0___

1.1.10.50 → Subnet Mask: ___255.0.0.0_____

220.90.130.45 → Subnet Mask: ___255.255.255.0___

134.125.34.9 → Subnet Mask: ___255.255.0.0___

95.250.91.99 → Subnet Mask: ___255.0.0.0___

Part 6: CIDR & Subnetting (10 points)

Convert and solve.

192.168.1.0/24 → Subnet Mask: __255.255.255.0_ Hosts: __254__

$32 - 24 = 8$ $256 - 2 = 254$

192.168.1.0/26 → Subnet Mask: __255.255.255.0__ Hosts: _62_

$32 - 26 = 6$ $64 - 2 = 62$

10.0.0.0/8 → Subnet Mask: _255.0.0.0__ Hosts: __16,777,216__

$32 - 8 = 24$

172.16.0.0/16 → Subnet Mask: __255.255.0.0__ Hosts: __65,534__

$32 - 16 = 16$

192.168.10.0/27 → Subnet Mask: _255.255.255.0__ Hosts: _30_

$32 - 27 = 5$ $32 - 2 = 30$

192.168.10.0/28 → Subnet Mask: __255.255.255.0_ Hosts: __14__

$32 - 28 = 4$ $16 - 2$